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March 15, 1966
INTERIM REVISION of
GGG-C-748b
August 21, 1959

INTERIM FEDERAL SPECIFICATION

CUTTER, CIRCLE (ADJUSTABLE);
FOR GASKETS AND WASHERS

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20406, based upon currently technical available information. It is recommended that Federal agencies use it in procurement and forward recommendations for changed to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GGG-C-748b, dated August 21, 1959.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers cutters (types I, II, and III) for cutting circular washers and gaskets of materials such as asbestos, rubber, canvas, cork, leather, felt, and cardboard. In addition type III, style B will cut irregular shapes and strips; and cutters (type IV) for cutting holes or disks out of sheet metal, brass, aluminum, fiber, plastic, and wood.

1.1.1 Federal specification coverage. Federal specifications do not include all the types and styles indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and styles. The cutters shall be of the following types and styles as specified (see 6.1):

Type I. Center pivoting.
Type II. End pivoting.
  Style A. Adjustable pivot holder, machine adapter, and bit shank.
  Style B. Nonadjustable pivot holder.
Type III. Block, movable pivoting.
  Style A. With extensions.
  Style B. Without extensions.
Type IV. Drill pivoting.
  Style A. Bit shank.
  Style B. Machine shank.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:
  GGG-D-751 - Drills, Twist (Taper-Square, Straight and Taper-Round Shanks).
  QQ-C-851 - Cutter Bit, Tool Holder (Solid High Speed Steel).

Federal Standards:
  Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
  Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:
MIL-H-15424 - Hand Tools; Packaging of.
MIL-G-12803 - Gasket Material, Non-metallic.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Governmental:
National Bureau of Standards Handbook:
H28 - Screw-Thread Standards for Federal Services, and Supplements thereto.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402)

3. REQUIREMENTS

3.1 Materials. Materials used in the construction of circle cutters shall be as hereinafter specified for each type and style. All forgings or castings shall be of uniform quality and free from defects and imperfections that may affect their serviceability.

3.1.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of cutters which are otherwise in accordance with this specification.

3.2 Graduations of linear measurement scale. Graduation lines, when applicable, shall be of a uniform and readily distinguishable width, straight, permanent, and free from discontinuities and ragged edges. If the graduation lines are on metal or plastic members, they shall be sunk below the surface of the member and shall have sharp edges. Graduation lines representing equal fractional aliquots of any scale shall be of equal height and readily distinguishable by their relative height from graduation lines representing other fractional aliquots of the same scale. Lines representing inches shall be marked with the corresponding figures. The scale error for any linear setting on the members shall not exceed 1/64 inch.

3.2.1 Figures and markings. Figures and markings shall be clear, distinct, and permanent. Figures and markings on cutters having linear measurement scales shall be applied in a manner identical with that used to apply the graduation lines.

3.3 Cutter blades. The cutter blades, except types III and IV (see 3.12.3 and 3.13.4), shall be of tool steel, hardened, and tempered from 40 to 52 on the Rockwell C scale (see 4.6.1). Cutter blades, except types III and IV shall have sharpened cutting tips to facilitate smooth, clean, accurate cutting actions. The blades and tool bits shall protrude at right angles (plus or minus 30 minutes) to the frame, beam, bar, or block, to insure correct setting and avoid chamfering the edges of the material being cut. Attachable cutter blades shall fit neatly in the holder with a minimum of play.

3.4 Cutter-blade holders. The cutter-blade holders, except for types III and IV, shall be suitably constructed of steel or malleable iron and securely and readily clampled to the crossbeam by means of knurled thumbscrews, nuts, or setscrews. Means shall be provided for attaching the cutter blade securely to the holder.

3.5 Pivot. The pivot, except for types III and IV, (see 3.12.4 and 3.13.3) shall be of tool steel, suitably hardened and tempered, cylindrical in cross section and of proper length and size. The pivot shall be provided with a gradual taper at the point to assure penetration of the various thicknesses of material. The pivot shall protrude at right angles (plus or minus 30 minutes) to the beam or frame.

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3.6 Nuts, thumbscrews, and setscrews. The nuts, thumbscrews, and setscrews shall be of steel, except for type III, which shall be provided with a threaded brass or steel insert, properly engaging the threads provided to lock the cutter blades or cutter blade holders to the beam, bar, or block. Screw threads shall be in accordance with Handbook H28.

3.7 Repair parts. The successful bidders supplying cutters covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.8 Marking. Each cutter shall be plainly and permanently marked with the manufacturer's name or trademark, readily identifying the source of manufacture.

3.9 Finish. The cutters shall be finished in accordance with the manufacturer's standard practice.

3.10 Type I, center pivoting. Type I cutters shall consist of a crossbeam, bitstock pivoting body, pivot, cutter blades and cutter-blade holders. Type I cutters shall be capable of withstanding the test specified in 4.6.3.1 and shall be similar to figures 1 or 2.

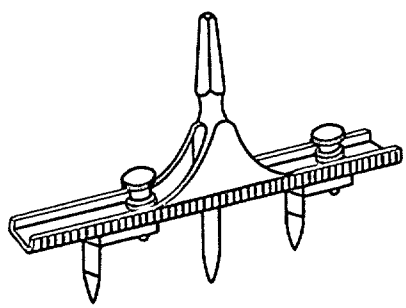


FIGURE 1. Type I, center pivoting, rectangular crossbeam.

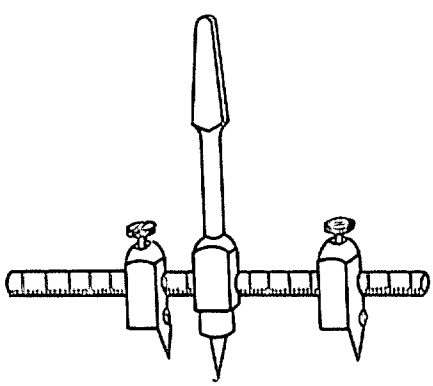


FIGURE 2. Type I, center pivoting, cylindrical crossbeam.

3.10.1 Crossbeam. The crossbeam shall be of steel or malleable iron, and graduated into increments of 8ths or 16ths of an inch. The numerals at the corresponding inch mark shall begin at zero at the center of the crossbeam with the numerals increasing toward the ends of the crossbeam.

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3.10.1.1 Rectangular crossbeam construction. When rectangular crossbeam construction is utilized, cutters shall have a rectangular crossbeam with two integral cutting blades, or two cutting blade holders with cutters (see figure 1). The crossbeam shall be designed to provide a suitable passage and movement of the thumbscrews for adjusting the cutter blades or cutter-blade holders to the desired setting.

3.10.1.2 Cylindrical crossbeam construction. When cylindrical crossbeam construction is utilized, cutters shall have a cylindrical crossbeam with two cutter blade holders and cutter blades (see figure 2). A flat surface shall be provided on the crossbeam to accommodate the ends of the thumbscrews and to position the holders in proper alignment.

3.10.2 Bitstock pivoting body. The bitstock pivoting body shall be of steel or malleable iron, and integral with or securely attached to the crossbeam. The tapered square bit shank shall have an included angle of 9 degrees, plus or minus 1 degree, and shall be approximately 7/16 inch square on the large end and approximately 1-1/4 inches long. The round stock directly below the tapered shank shall be 3/8 inch in diameter by a minimum of 3/4 inch long to facilitate a satisfactory fit and be held correctly in bitstock.

3.10.3 Pivot. The pivot shall be firmly and rigidly secured by a setscrew, chuck, or forced fit, and shall be in accordance with 3.5. Pivots having a forced fit shall be capable of being removed for replacement or resharpening.

3.10.4 Capacity. Type I cutters shall cut material up to and including 1/8 inch in thickness and from 3/4 to, and including, 6 inch outside diameters and 5/8 to, and including, 5-1/2 inch inside diameters.

3.10.5 Cutter blades. For cutters having a rectangular crossbeam the cutter blades shall be integral with the holder or attachable to the cutter holder. For cutters having a cylindrical crossbeam, the cutter blades shall be securely attached to the holders and be in alignment with the pivot. All cutter blades shall be in accordance with 3.3.

3.10.5.1 Cutter-blade holders. All cutter-blade holders shall be in accordance with 3.4. The hole provided in the cutter holders for the cylindrical crossbeam shall fit the crossbeam without binding or excessive play.

3.11 Type II, end pivoting. Type II cutters shall be of steel and consist of a beam, pivot holder, pivot, cutter blade(s) and cutter-blade holder(s). The cutters shall be capable of withstanding the test specified in 4.6.3.1.

3.11.1 Pivots. The pivots shall conform to 3.5 and be firmly and rigidly secured by a setscrew, chuck, or forced fit.

3.11.2 Style A, adjustable pivot holder, machine adapter, and bit shank. Style A cutters shall have a square beam, two cutter blades, and pivot holder.

3.11.2.1 Beam. One end of the beam shall be designed to provide a suitable passage and movement of the cutter-blade holders to the desired setting.

3.11.2.2 Pivot holder. The pivot holder shall be suitably constructed of a machine adapter shank, bit shank, and body. The machine adapter and bit shank shall be so constructed as to assure a satisfactory fit and be held correctly in a drill press or a hand bit brace. Means shall be provided in the body for securely attaching the holder to the beam and the pivot to the holder. The hole provided in the holder shall fit the beam without binding or excessive end play.

3.11.2.3 Cutter blades. The cutter blades shall be securely attached in the holders and in alignment with the pivot. The cutter blades shall be in accordance with 3.3.

3.11.2.4 Cutter-blade holders. The cutter-blade holders shall be in accordance with 3.4.

3.11.2.5 Capacity. The cutter shall cut materials up to and including 1/8 inch in thickness and from 1-inch up to, and including, 24-inch diameters.

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3.11.2.6 Type II, style A cutters shall be similar to figure 3.

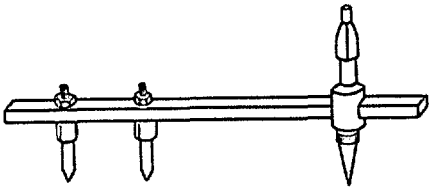


FIGURE 3. Type II, style A, end pivoting, adjustable pivot holder, machine adapter and bit shank.

3.11.3 Style B, nonadjustable pivot holder. Style B cutters shall have a rectangular beam, two cutter blades or cutter-blade holders, end pivot, and handles.

3.11.3.1 Beam. The beam shall be graduated into increments of 8ths or 16ths of an inch. The numeral at the corresponding inch mark shall begin at the center of the end pivot with the numerals increasing toward the opposite end of the beam. An elongated slot of suitable width and length shall be provided for passage and movement of the thumbscrews and for adjusting and securing the cutter blades or cutter holders to the desired setting. Means shall be provided at one end of the beam, to permanently secure the end pivot.

3.11.3.2 Handles. Handles shall be of clear, straight-grained hardwood. Handles shall be round or suitably designed to facilitate a comfortable and secure handgrip. All exposed wooden surfaces shall be protected with a durable coating of enamel, varnish, shellac, or lacquer. Means shall be provided for attaching one handle permanently to the beam at the pivot end and the other handle by a thumbscrew to the outer cutter-blade holder.

3.11.3.3 Cutter blades. The cutter blades shall be integral with holder or attachable to cutter-blade holder and in alignment with the pivot. Attachable cutter blades shall be in accordance with 3.3.

3.11.3.4 Cutter-blade holders. The cutter-blade holders shall be in accordance with 3.4.

3.11.3.5 Capacity. The cutter shall cut materials up to, and including, 1/8 inch in thickness and from 4 inches up to, and including, 20 inch diameters.

3.11.3.6 Type II, style B cutters, shall be similar to figure 4.

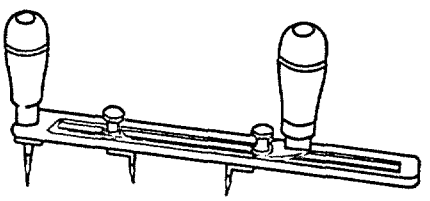


FIGURE 4. Type II, style B, end pivoting, nonadjustable pivot holder.

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3.12 Type III, block, movable pivoting. Type II cutters shall consist of a block, pivot post with nut and washer(s), cutter blades, pivot pins, and cutting board with centering ferrule(s). A metal carrying case shall be provided to accommodate the cutter and its components, except the cutting board. Type III cutters shall be capable of withstanding the test specified in 4.6.3.2.

3.12.1 Block. The block shall be either metal or plastic with an elongated slot of suitable depth, width, and length to provide for housing, supporting and adjusting the pivot post to the desired setting. Suitable means shall be provided for locating and securing cutter blades in the end of the block. The top or bottom of the block, along the side of the pivot post slot, shall be graduated in 1/32 inch increments, that, when set, will equal the actual diameter desired. The opposite side of the pivot post slot may be graduated in millimeters. The numerals at the marks shall start with zero at the cutter blade end of the pivot post slot, with the numerals increasing toward the opposite end. The graduations and numerals may be either sunk into the block itself, or etched or sunk below the surface of a plate securely fastened to the block. They shall otherwise conform to 3.2.

3.12.2 Pivot post. The pivot post shall consist of a metal or plastic body with a ferrule (properly inserted for centering the pivot pins), a stud screw permanently attached, fiber washer(s), and a suitable knurled nut for securing the pivot post to the block.

3.12.3 Cutter blades. The cutter blades shall be properly formed to a sharp edge to facilitate a smooth and accurate cutting action. The blades shall be of high carbon steel, hardened and tempered to a hardness of from 90 to 93 on the Rockwell superficial 15N scale under a 15 kilogram (kg) load (see 4.6.1).

3.12.4 Pivot pins. The pivot pins shall be of corrosion resisting steel or tool steel suitably hardened and tempered, cylindrical in cross section, of proper length and size. The pivot pin shall be provided with a gradual taper at the point to assure penetration of the various thicknesses of material, except for the nonpiercing pivot pin (see 3.12.7 and 3.12.8). The pivot pin shall protrude at right angles, plus or minus 30 minutes, to the base of the block.

3.12.5 Cutting board. The cutting board shall have a plastic cutting surface, with a durometer hardness of 90 plus or minus 5 (see 4.6.2), with a suitable rigid backing for support and corrosion-resisting steel centering ferrule(s) inserted. The cutting surface shall be rough in order that gaskets will not slip while being cut. The plastic pad shall be removable and replaceable. The ferrule(s) shall be serrated to preclude turning or moving in the board and shall be perpendicular to the surface within plus or minus 30 minutes. Holes in ferrules shall fit pivot pins neatly or as required for a close running fit.

3.12.6 Carrying case. A metal carrying case shall be provided to suitably accommodate the tool and components, except the cutting board. The case shall be of sheet metal of durable and rigid construction. The case shall be equipped with a hinged cover and a latching device. An instruction sheet containing clear and concise information as to the proper operation of the tool shall be included inside the carrying case. A parts list including manufacturer's replacement parts and identification numbers shall be permanently attached to the top of the case on the inside.

3.12.7 Style A, with extensions. Style A cutters shall be provided with three each of 0.017 and 0.025 inch thickness steel blades, two extension arms, three varied length pivot pins, one nonpiercing pivot pin for cutting solid discs or diaphragms, one pin handle for seating pin in material, one hexagon key wrench to tighten pivot post nut, one pivot post with two washers and a nut with a hexagon wrench hole, and a cutting board with two centering ferrules. The blade shall be held firmly by a plate and one screw with a knurled knob-type head.

3.12.7.1 Capacity. The cutter shall be capable of cutting by successive cuts of at least 1/8 inch deep, when necessary, through pliable conventional gasket materials from 0.001 through 0.750 inch in thickness. Thick materials shall be cut through by cutting with the blade partially extended. Subsequent cuts are made after extending the blade farther until the blade cuts through 0.750 inch thick material. Style A cutters shall be capable of cutting diameters from 1 to 13 inches without extension arms and with extension arms added up to, and including, 36 inches in diameter. For diameters under 2 inches, the thickness of of the material cannot exceed one-half the diameter in order to clear the blade.

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3.12.7.2 Body. The overall length of the body with the cutter head attached shall be not less than 7 inches. The width shall be not less than 1-11/16 inches.

3.12.7.3 Cutter blades. The blades shall be provided with sharpened cutting tips to facilitate a smooth and accurate cutting action. The blades shall otherwise conform to 3.12.3.

3.12.7.4 Extension arms. Each extension arm shall be capable of adding an extra 12 inches to the diameter. They shall be interchangeable and capable of fitting onto the pivot post in the cutter body. They shall be capable of holding the pivot pin steady and true. Suitable means shall be provided to keep the extension arms in line and keep them from bending.

3.12.7.5 Cutting board. The cutting board shall be at least 18 by 18 inches square and shall have two centering ferrules, one in the center for cutting diameters up to 18 inches and one ferrule in a corner for cutting larger diameters. The board shall otherwise conform to 3.12.5.

3.12.7.6 Type III, style A cutters shall be similar to figure 5.

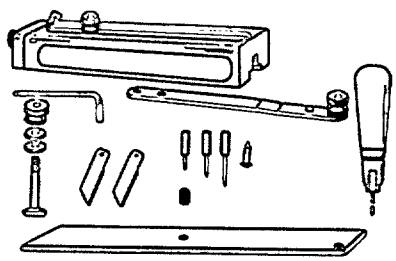


FIGURE 5. Type III, style A, block, movable pivoting, with extensions.

3.12.8 Style B, without extensions. Style B cutters shall be capable of following irregular lines and cutting circles, odd shapes, and strips. The cutter shall be furnished with six cutting blades (four pointed and two curved), six pivot pins (two short, two long, and two nonpiercing pivot pins for cutting discs), a pivot post with nut and washer, and a cutting board with one centering ferrule. The blade shall be held firmly by a plate and one screw with a knurled knob-type head.

3.12.8.1 Capacity. The cutter shall be capable of cutting, by successive cuts of at least 1/8 inch deep when necessary, through pliable, conventional, gasket materials from 0.001 through 0.375 inch in thickness. Thick materials shall be cut through by cutting with the blade partially extended. Subsequent cuts are made after extending the blade farther until the blade cuts through 0.375 inch thick material. Style B cutters shall be capable of cutting diameters from 1/4 inch to 6 inches. For diameters under 1 inch, the thickness of the material cannot exceed one-half the diameter in order to clear the blade.

3.12.8.2 Body. The overall length of the body shall be not less than 4-7/16 inches. The width shall be not less than 1-1/4 inches. The body shall have a tracing index line aligned with the blade for following irregular lines.

3.12.8.3 Cutting board. The cutting board shall be at least 6 by 6 inches square and shall have one centering ferrule. The board shall otherwise conform to 3.12.5.

3.12.8.4 Cutter blades. The cutter blades shall conform to 3.12.3 and be true within a tolerance of plus or minus 1/64 inch to the tracing index line on the block in order to follow irregular cutting lines.

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3.12.8.5 Type III, style B cutters shall be similar to figure 6.

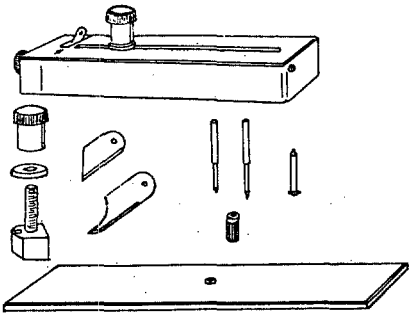


FIGURE 6. Type III, style B, block, movable pivoting, without extensions.

3.13 Type IV, drill pivoting. Type IV cutters shall consist essentially of an adjustable bar, body, drill, and tool bit.

3.13.1 Adjustable bar. The adjustable bar shall be of good-quality steel properly heat treated to resist wear or deformation. The bar shall be either circular, square, or rectangular in cross section. The circular bar shall be provided with a suitable flat surface of sufficient length to accommodate the end of the holding screw for securing and positioning the bar. Suitable means shall be provided at one end of the bar to properly accommodate and secure the tool bit parallel to the surface of the flat and at a right angle to the longitudinal axis of the bar. The tool bit shall not become loose or detached under normal service conditions.

3.13.2 Body. The body shall be of good quality steel properly heat-treated to resist wear or deformation. The body shall be provided with a smooth hole or channel at a right angle to the vertical axis for passage of the bar, and a smooth hole concentric with the axis of the body to accommodate the drill. Suitable means shall be provided in the body for locking the bar and drill. The bar and drill shall not become loose or detached under normal service conditions.

3.13.4 Tool bit. The tool bit shall be in accordance with type I, style 1, grade A, of 00-C-851, except tool bits shall have an overall length of 2-1/4 inches minimum when cutting end is finished ground. Type IV, style A cutters shall be provided with a 3/16 inch tool bit. Type IV, style B cutters shall be provided with a 1/4-inch tool bit. Each bit shall be sharpened in accordance with the manufacturer's standard practice, and the bit may be otherwise modified to fit the bit holder at the manufacturer's option.

3.13.5 Style A, bit shank. Style A cutters shall have a tapered squared bit shank suitable for use in a hand bit brace. The shank shall have an included angle of 9 degrees, plus or minus 1 degree, and shall be approximately 7/16 inch square on the large end and approximately 1-1/4 inches long. Style A cutters shall be capable of withstanding the test specified in 4.6.3.3.

3.13.5.1 Drill. The drill shall be 1/4 inch in diameter with a straight shank, and be furnished in a length suitable for the purpose intended. Except for length, the drill shall be in accordance with the requirements and tests in GGG-D-751.

3.13.5.2 Capacity. Style A cutters shall be capable of cutting up to 1/16 inch thick brass and aluminum, sheet fiber, plastic, or wood up to 3/8 inch in thickness. The cutters shall be capable of cutting from 1 inch up to, and including, 5-1/2 inch diameters.

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3.13.5.3 Type IV, style A cutters shall be similar to figure 7.

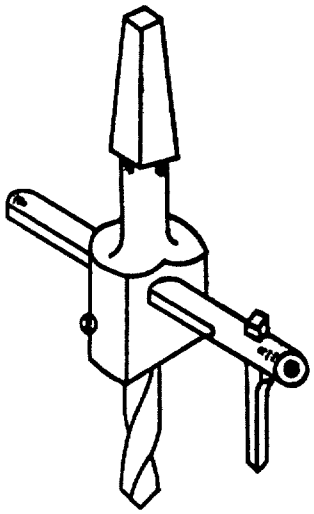


FIGURE 7. Type IV, style A, drill pivoting, bit shank.

3.13.6 Style B, machine shank. Style B cutters shall have a 1/2 inch straight shank. Style B cutters shall be capable of withstanding the test specified in 4.6.3.4.

3.13.6.1 Drill. The drill shall be either 1/4 or 3/8 inch in diameter with a straight shank, and be furnished in a length suitable for the purpose intended. The drill shall be in accordance with the requirements and tests in GGG-D-751.

3.13.6.2 Capacity. Style B cutters shall be capable of cutting sheet steel up to 10-gage (0.1345-inch), brass and aluminum, sheet fiber, plastic, and wood up to 3/4 inch in thickness. The cutter shall be capable of cutting from 1-7/8 inches up to, and including, 8 inches in diameter.

3.13.6.3 Type IV, style B shall be similar to figure 8.

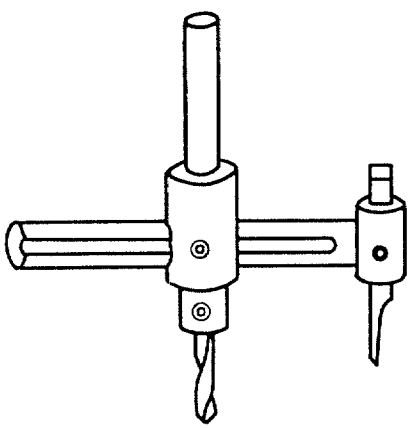


FIGURE 8. Type IV, style B, drill pivoting, machine shank.

3.14 Workmanship. Workmanship shall be equal to the highest grade, and in accordance with good commercial practices. Rust, fins, burrs, or any imperfections which may impair the serviceability or durability of cutters are not acceptable.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table I.

Table I. Sampling data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual and dimensional examination	1 ea.	II	4.0	Defects per hundred units	4.3.1
Testing	1 ea.	S-3	1.0	Percent defective	4.5
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.7

4.3 Examination.

4.3.1 Visual and dimensional examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, marking, and dimensions.

4.4 Classification of defects. Inspection of the cutters shall be conducted as follows:

4.4.1 Visual and dimensional examination.

Categories and Defects

AQL (approx.) 1.5 percent

100. Critical:
None defined

200. Major (sampling per table I).
200.1 Type and style not as specified.
200.2 Incomplete, component parts missing.
200.3 Materials not as specified, not free of cracks, splits, or breaks.
200.4 Evidence of rust, fins, burrs, sharp or rough edges.
200.5 Graduations not uniform and readily distinguishable, not straight and permanent, or graduations not as specified. (except style A, type II and type IV).
200.6 Scale error exceeds the maximum allowable, (except type IV).
200.7 Bitstock pivoting body defective, not integral with (or securely attached to) the cross-beam (type I only).
200.8 Pivot defective, not of the proper length, not tapered, not firmly and rigidly secured, or not capable of being removed for replacement or resharpening.
200.9 Pivot holder defective, machine adapter and bit shank not as specified, holder not adjustable along the beam, holder binds or evidences excessive play, or cannot be locked in any position along the beam (type II only, except style B).
200.10 Pivot post defective, stud screw not securely attached or threads damaged, ferrule missing or improperly inserted, or nut fails to allow clearance for the stud screw when securing pivot post to block (type III only).

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200.11 Pivot not of the proper length, or not as specified (type III only).
200.12 Cutter blades not sharp, not rigidly attached, or not as specified (type II only).
200.13 Cutter blades defective, not sharp, not rigidly attached, or not as specified (type I only).
200.14 Cutter blades defective, not pointed, not sharp, or evidence of excessive play when cutter blades are attached to the box (type III only, style B).
200.15 Cutter blade holders defective, cannot be readily clamped to the crossbeam, clamping screwhead (or nut) not knurled, or cutter blade holders stick or bind when moved along the beam (types I and II only).
200.16 Cutter blades defective, not formed as specified, not sharp, or evidence of excessive play when cutter blades are attached to the box (except style A, type III and type IV).
200.17 Handles not shaped as required, not coated, pivot handle not permanently attached, or other handle not attached by a thumbscrew (type II only, except style A).
200.18 Capacity of cutter not as required.
200.19 Angle of cutter blades or pivot with beam not within the specified tolerance (except type IV).
200.20 Body defective, cannot be locked to the bar, body cannot be moved along the bar, or evidence of sticking, binding, or excessive looseness, hole or channel not right angle to vertical axis (type IV only).
200.21 Drill not sharp, not the required length and size, or drill not as specified (type IV only).
200.22 Tool bit not sharp, not the required size, not as specified, or tool bit not securely attached to bar (type IV only).
200.23 Size of machine shank not as specified (type IV only, except style A).

300. Minor (sampling per table I):
300.1 Marking, manufacturer's name or trademark missing, illegible, incorrect, or not permanent.
300.2 Tapered square bit shank dimensions not as required (types I and IV only).
300.3 Length of round stock (below tapered shank) not as specified (type I only).

4.5 Tests. Each of the sample cutters or cutting boards selected in accordance with 4.3 shall be subjected to the hardness and performance tests specified in 4.6.1 through 4.6.3.4, as applicable to verify compliance with this specification. Any sample cutter or cutting board which does not conform to these characteristics shall cause rejection of the lot represented by the sample.

4.5.1 Acceptance tests (possible tests failures).

(a) Hardness:
(1) Hardness of cutter blades not within the specified values on the Rockwell C scale (types I, II, and III only).
(2) Hardness of drill less than the specified minimum values on the Rockwell C scale (type IV only).

(b) Performance test:
(1) Failure of cutter to cut the required number of washers, circles not concentric, not the same size, or cuts not clean and positive when cutting the specified material, cutter breaks, deforms, chips, or cracks (types I, II, and III only).
(2) Failure of cutter to cut the required number of disks, disks not concentric, not the same size, or cuts not clean and positive when cutting the specified material, cutter breaks, deforms, chips or cracks (types IV only).

4.6 Test procedures.

4.6.1 Hardness test. Each sample cutter selected from each inspection lot in accordance with 4.3 and the cutter blades shall be subjected to a hardness test in accordance with method 243 of Fed. Test Method Std. No. 151 to determine conformance with 3.3, 3.12.3, 3.13.5.1, and 3.13.6.1.

4.6.2 Durometer hardness test for cutting board. Each sample cutting board selected from each inspection lot in accordance with 4.3 shall be subjected to a durometer hardness test to determine conformance with 3.12.5.

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4.6.3 Performance test. Each sample cutter selected from each inspection lot in accordance with 4.3 shall be subjected to the applicable test specified in 4.6.3.1, 4.6.3.2, 4.6.3.3, and 4.6.3.4.

4.6.3.1 Types I and II. Each sample cutter shall be tested by cutting 10 washers of the same diameter from either 1/16 inch thickness, type I, (P1161-A) gasket material under MIL-G-12803 or stiff cowhide leather of 1/8 inch thickness. The sample cutters shall consistently cut concentric circles identical in size within plus or minus 1/32 inch, and show a clean and positive cut (see 3.3).

4.6.3.2 Type III. Each sample cutter shall be tested by cutting 10 washers of the same diameter from 1/8 inch thick sheet asbestos. The blade shall be extended sufficiently to cut entirely through the test specimen on the first cut. The sample cutters shall consistently cut concentric circles identical in size within plus or minus 1/32 inch, and show a clean and positive cut (see 3.3 and 3.12).

4.6.3.3 Type IV, style A. Each sample cutter shall be tested by cutting 10 disks of the same diameter out of 16-gage (0.0598-inch) mild sheet steel. The sample cutters shall consistently cut concentric disks identical in size within a plus or minus 1/64 inch, and show a clean and positive cut (see 3.13.5).

4.6.3.4 Type IV, style B. Each sample cutter shall be tested by cutting 10 disks of the same diameter out of 10-gage (0.1345 inch) sheet steel. Sample cutters shall consistently cut concentric disks identical in size within plus or minus 0.005 inch, and show a clean and positive cut (3.13.6).

4.7 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Table II. Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), the cutters shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing, A, B, or C, as specified (see 6.1).

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and style required (see 1.2).

(c) Selection of applicable level of preservation and packaging and the level of packing required (see 5.1).

6.1.1 When repair parts for cutters are required, the following should be specified:

(a) Manufacturer's name or trademark of cutter for which required.

(b) Name of part.

(c) Quantity of each item.

(d) Any other data applicable to cutter which will assist in securing the proper repair parts.

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